



CENTRAL UNIVERSITY OF SOUTH BIHAR

Department of Computer Science

Class Test-I, 14th March. 2026

Subject: DBMS

Course Code: CSC82DC00604

Semester: Ist

Time: 1 Hour

M. M.: 10

Session: 2025-2026

1. Define: (A) DBMS (B) Data Independence. (1+2=3)
2. Explain data models and its types. (4)
3. Write Short notes on (any two): (3)
 - (A) DML & DDL
 - (B) Schema and an Instance
 - (C) Can you perform calculations within a SELECT statement?



CENTRAL UNIVERSITY OF SOUTH BIHAR

Department of Computer Science

Class Test-II, 16th April 2026

Subject: DBMS

Course Code: CSC82DC00601

Semester: II

Time: 1 Hour

M. M.: 10

Session: 2025-2026

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1. What is logical database design? (1)
 2. What is normalization in DBMS? (1)
 3. What is an Entity–Relationship (ER) model in DBMS design? List out the relationships a conceptual design of an ER diagram for a University Database System. (3)
 4. Write an SQL query to display students who live in Patna. (2)

(Assume a table *Student (StudentID, Name, Course, Age, City)* in Database Management System.)

5. Describe the relationship between **super key**, **candidate key**, and **primary key**. (3)



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Question Paper of End –Term Examination, May 2026

Name of Department: Computer Science

Programme: M.Sc.(CS)

Session:2025-27

Semester: II

Course Title: Database Management System

Course Code: CSC82DC00604

Course Credit: 4

Max. Marks:70

Duration: 3:00Hrs.

Enrolment No. (to be filled by Student):.....

Instruction:

1. Attempt questions in all the Sections as directed.

Section - A

Answer *all* questions from below, each carrying 1 marks.

(10 X 1 = 10)

1. What is DBMS?
2. List out the functionalities of DBMS.
3. What is the role of Database Administrator?
4. What is data dictionary?
5. What is E-R model?
6. Define SQL.
7. What are the ACID properties?
8. Why Abstraction is needed?
9. What is Relational Algebra?
10. What is procedural DML?

Section - B

Answer any *five* questions below, each carrying 6 marks.

(5 X 6 = 30)

1. What is the need of the normalization? What is 3NF and BCNF?
2. Draw and explain an E-R diagram for a banking enterprise with almost all components.
3. Describe the three levels of data abstraction.
4. What are the steps in designing a database? Explain.
5. Describe the DDL and DML commands used in SQL with syntax and examples.
6. What is Data Model? Define the different type of Data models.

Section - C

Answer any two questions below, each carrying 15 marks.

(2 X 15 = 30)

1. Consider a table **STUDENT** with the following attributes:

(3+3+3+3+3)

(Student_ID, Name, Department, Marks)

Write SQL query to:

- a) Insert two records into the table.
 - b) Update the marks of a student whose Student_ID = 101.
 - c) Delete the record of a student whose Student_ID = 102.
 - d) Add a new column **Email** to the **STUDENT** table.
 - e) Grant **SELECT** permission on the **STUDENT** table to a user named *user1*.
2. What is a Database system? Explain the purpose of database systems. What are the advantages of DBMS over File System?

(2+8+5)

3. Describe the following:

(3+3+3+3+3)

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|--------------------------|--------------------------------|--------------------|
| a) Instances and Schemas | b) Data Independence | c) Serializability |
| d) Integrity Constraints | e) Primary key and Foreign key | |
